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Semi-prepared Airfield Condition Index (SPACI)

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Outline

- Steps to Inspect / Identify distresses
- Surface Conditions (C-17, C130)
- Distress Types
- Calculate the Results
- Exercise
- Nontraditionally Surfaced Airfields

Step 1: Divide the airfield into sections

- Inspect the airfield
- Divide the entire field into features and sections.
- Each section should be 250 feet long and the width of the runway or taxiway. The hammerheads and overruns should each be a section
- As a minimum for contingency operations, sections should be located in the touchdown area, primary braking area, point of aircraft rotation and the last 500 feet of the runway

All information pulled from ETL 97-9 and AFCEC

Step 2: Inspect Airfield And Identify Stresses

- Conduct as detailed and accurate an inspection as time and conditions permit
- There are eight possible problems or distresses
  - Potholes, Ruts, Loose aggregate size, Loose aggregate coverage (added), Dust, Rolling resistant material, Jet blast erosion, and Stabilized layer failure
- In each section you may have no distresses or multiple
- If unable to determine the type of distress, make a reasonable guess

Surface Condition
C-17 Semi-prepared Surfaces

Distress Severity Levels for C-17 Semi-prepared Surfaces

Distress Types	GREEN	AMBER	RED
91 Potholes	4” Deep and/or <15” Diameter	4” to 9” Deep and >15” Diameter	> 9” Deep and >15” Diameter
92 Ruts	Exist but < 4” Deep	4” to 9” Deep	>9” Deep
93 Loose Agg. Coverage	Covers < 1/10 of section	Covers between 1/10 and 1/2 of section	Covers > 1/2 of section
93 Loose Agg. Size	Max. ¾”, Recommended ½”	Max. ¾” to 1”	Max. >1”
94 Dust	Does not obstruct visibility	Partially obstructs visibility	Thick, obstructs visibility
95 Rolling Resistant Mat.	Exist but < 3.5” Deep	3.5” to 7.75” Deep	> 7.75” Deep
96 Jet Blast Erosion	Exist but < 1” Deep	1” to 3” Deep	> 3” Deep
97 Stabilized Layer Failure	Exist but < 1” Deep	1” to 2” Deep	> 2” Deep

Note: Potholes, ruts, and rolling resistant material are considered major distresses. Depending upon actual distress location, any of these distress types categorized as RED may cause overall airfield condition to be RED.

Training sites must meet green criteria, Contingency sites must meet yellow criteria

Surface Condition
C-130 Semi-prepared Surfaces

Distress Severity Levels for C-130 Semi-prepared Surfaces

Distress Types	GREEN	AMBER	RED
91 Potholes	4" Deep and/or <15" Diameter	4" to 6" Deep and >15" Diameter	> 6" Deep and >15" Diameter
92 Ruts	Exist but < 4" Deep	4" to 6" Deep	>6" Deep
93 Loose Agg. Coverage	Covers < 1/10 of section	Covers between 1/10 and 1/2 of section	Covers > 1/2 of section
93 Loose Agg. Size	Max. ¾", Recommended ½"	Max. ¾" to 1"	Max. >1"
94 Dust	Does not obstruct visibility	Partially obstructs visibility	Thick, obstructs visibility
95 Rolling Resistant Mat.	Exist but < 1" Deep	1" to 3" Deep	> 3" Deep
96 Jet Blast Erosion	Exist but < 1" Deep	1" to 3" Deep	> 3" Deep
97 Stabilized Layer Failure	Exist but < 1" Deep	1" to 2" Deep	> 2" Deep

Note: Potholes, ruts, and rolling resistant material are considered major distresses. Depending upon actual distress location, any of these distress types categorized as RED may cause overall airfield condition to be RED.

Training sites must meet green criteria, Contingency sites must meet yellow criteria

Surface Condition
C-146 and LTFW Semi-prepared Surfaces

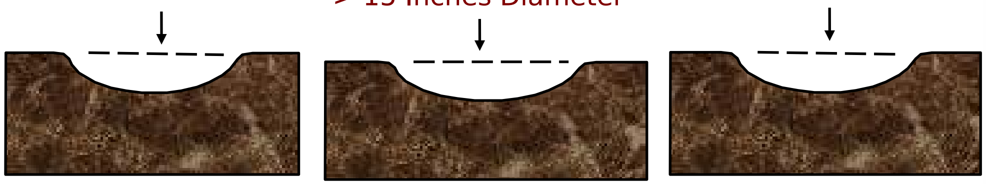
DISTRESS TYPES		GREEN C-146	AMBER C-146	RED C-146
91	Potholes	3" Deep and/or <15" Diameter	3" - 5" Deep and >15" Diameter	> 5" Deep and/or > 15" Diameter
92	Ruts (Main Gear)	Exist but < 3" Deep	3" - 5" Deep	> 5" Deep
92	Ruts (Nose Wheel)	Exist but < 1" Deep	1"-2" Deep	> 2" Deep
93	Loose Agg Coverage	Covers < 1/10 of Section	Covers between 1/10 & 1/2 of section	Covers > 1/2 of Section
93	**Loose Agg Size**	**Max 3/4", Recommended 1/2"**	**Max 3/4" - 1"**	**Max > 1"**
94	Dust	Does not obstruct visiblity	Partially Obstructs Visibility obstruct visiblity	Thick, Obscures visibility obstruct visiblity
95	Rolling Resistant Mat	Exist but < 1" Deep	1" to 2" Deep	> 2" Deep
96	Jet Blast Erosion	Exist but < 3/4" Deep	1" Deep	> 1" Deep
97	Stabilized Layer Failure	Exist but < 3/4" Deep	1" Deep	> 1" Deep

Note: Potholes, ruts, and rolling resistant material are considered major distresses. Depending upon actual distress location, any of these distress types categorized as RED may cause overall airfield condition to be RED. Training sites must meet green criteria, Contingency sites must meet yellow criteria.

91. Potholes

- Potholes are bowl-shaped depressions in the surface.
- Once formed, they will continue to disintegrate.
- If potholes have hard, abrupt, vertical sides, refer to stabilized layer failure criteria.
- Measure the depth of the biggest potholes and determine their severity level.

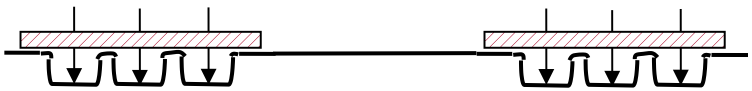
	GREEN	AMBER	RED
C-17	< 4 Inches Deep < 15 Inches Diameter	4 to 9 Inches Deep > 15 Inches Diameter	> 9 Inches Deep > 15 Inches Diameter
C-130	< 4 Inches Deep < 15 Inches Diameter	4 to 6 Inches Deep > 15 Inches Diameter	> 6 Inches Deep > 15 Inches Diameter



92. Ruts

- Ruts are surface depressions in the wheel paths that run parallel to the centerline. They result from repeated aircraft passes and become more severe with time.

- Check Rut Depths at touchdown, braking area, turn around, and point of rotation



- Lay a straightedge across the ruts with both ends resting on the solid runway surface with loose (rolling resistant material) removed. Measure the average depth of the three deepest ruts on each side, from the bottom of the straightedge to the solid ground in the bottom of the rut. Use the higher average(left or right) depth for that location. Rut width does not affect severity.

Location 1 Measurements	Location 2 Measurements	Location 3 Measurements	Location 4 Measurements
1. ___ 4. ___	1. ___ 4. ___	1. ___ 4. ___	1. ___ 4. ___
2. ___ 5. ___	2. ___ 5. ___	2. ___ 5. ___	2. ___ 5. ___
3. ___ 6. ___	3. ___ 6. ___	3. ___ 6. ___	3. ___ 6. ___
Loc 1 Max= ___	Loc 2 Max= ___	Loc 3 Max= ___	Loc 4 Max= ___

Runway Maximum Rut Depth = ____

Rut Depths	Operational Code	Operational Code Description
>9" (C-17) >6" (C-130)	**Red**	**Operate Aircraft Only in Emergency Situations.**
4" – 9" (C-17) 4" – 6" (C-130)	**Amber**	**Operate Aircraft for a Limited Number of Operations.**
Exist but < 4"	**Green**	**Operate Aircraft for Unlimited Mission Operations.**

93. Loose Aggregate Coverage

- Loose aggregate consists of small stones ¼-inch or larger that have separated from the soil binder.
- In large enough quantities and sizes, it becomes dangerous.
- Rocks over 4 inches in diameter must be removed from the airfield before any operations.
- If material crushes underfoot, it is not considered loose aggregate
- Estimate the area of the section that is covered by loose aggregate assess appropriate severity

Maximum Aggregate Coverage	Operational Code	Operational Code Description
Covers > 1/2 of section	**Red**	**Operate Aircraft Only in Emergency Situations.**
Covers between 1/10 and 1/2 of the section	**Amber**	**Operate Aircraft for a Limited Number of Operations.**
Covers < 1/10 of section	**Green**	**Operate Aircraft for Unlimited Mission Operations.**

93. Loose Aggregate Size

- ¾" to 1" creates engine FOD potential, as well as probable damage to undercarriage antennae
- Larger aggregate provides tire cutting potential as well as greater potential for undercarriage FOD damage
- Recommend surface coarse gradation with maximum ½" aggregate size be used on LZs

Maximum Aggregate Size	Operational Code	Operational Code Description
> 1"	**Red**	**Operate Aircraft Only in Emergency Situations.**
3/4" - 1"	**Amber**	**Operate Aircraft for a Limited Number of Operations.**
< 3/4"	**Green**	**Operate Aircraft for Unlimited Mission Operations.**

Step 3: Calculating the Ratings

SPACI Example:

Distress Type	Severity	Deduct Value
91		
92		18
93		4
94		4
95		22
96		
97		

Total Deduct Value = 48

q = 2

SPACI = 65

Note: If any distress is red, the landing zone safety officer will determine the suitability for operations.

97	Stabilized Layer Failur	5	15	10	25	15	35

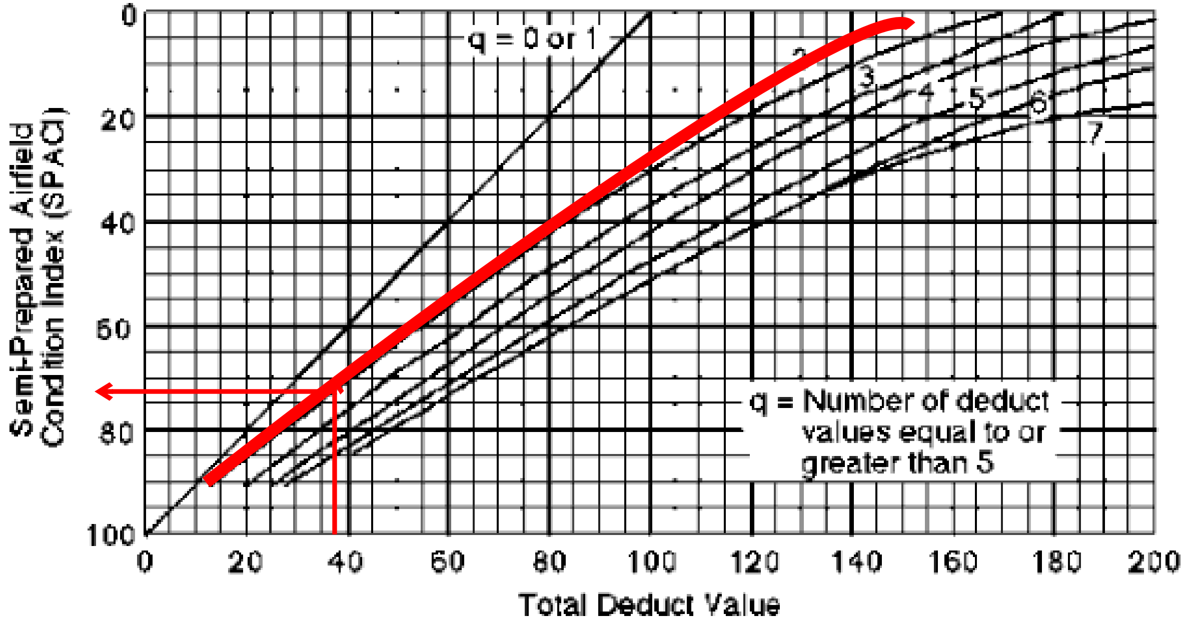
*RT - Runways & Taxiways

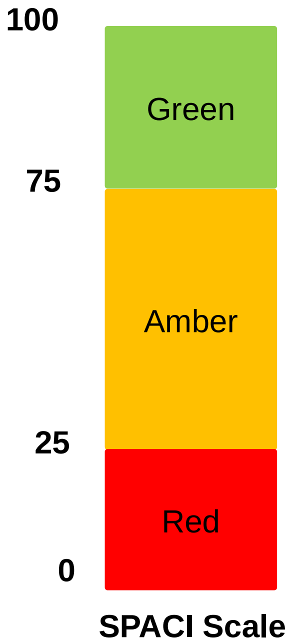
**H/A - Hammerheads & Aprons

NOTE: 247 is the highest deduct value possible (HA** all Red)

Step 3: Calculating the Ratings

- On the SPACI curve locate the “Total Deduct Value.”
- Utilize the numbers on the bottom of the graph until you reach the total deduct value of 36.
- Go up the vertical line until it intersects with the curved line that matches your *q* number (in this example, 2). From that point, follow the line across to the left to find the rating (the SPACI). For this example, the SPACI is 73.
- Go to the SPACI scale which shows that 73 is in the Amber range.
- Circle this overall rating on the bottom left of the inspection sheet.





SPACI EXERCISE

C-17 operations:

3’’ deep and 12’’ diameter potholes

3’’ ruts

Partially obstructed dust

1.5’’ deep Rolling Resistant Material

4’’ deep Jet Blast Erosion

1.5’’ deep Stabilized Layer Failure

C-130 operations:

3.5’’ deep and 13’’ diameter potholes

Partially obstructed dust

5’’ jet blast erosion

1.5’’ stabilized layer failure

Results:

Deduct Value=65 / q=4 / SPACI=61 / Amber

Deduct Value=33 / q=2 / SPACI=76 / Green

Questions / Critiques

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